gas generator, valve actuator and other control system element characteristics in addition to the fan response. The critical response time constant is the 0.3 second τ_{sys} required for an upward flight path control command during normal operations. Special studies have indicated it is feasible through the use of small amounts of symmetrical thrust spoiling during neutral control and other control system quickening techniques to meet the critical system flight path control time constant requirement with fan time constants of the order of 0.3 seconds. Figure 42 presents estimated fan alone time constants, τ_{fan}, for a range of sizes of 6 and 8 fan systems. The figure shows that the

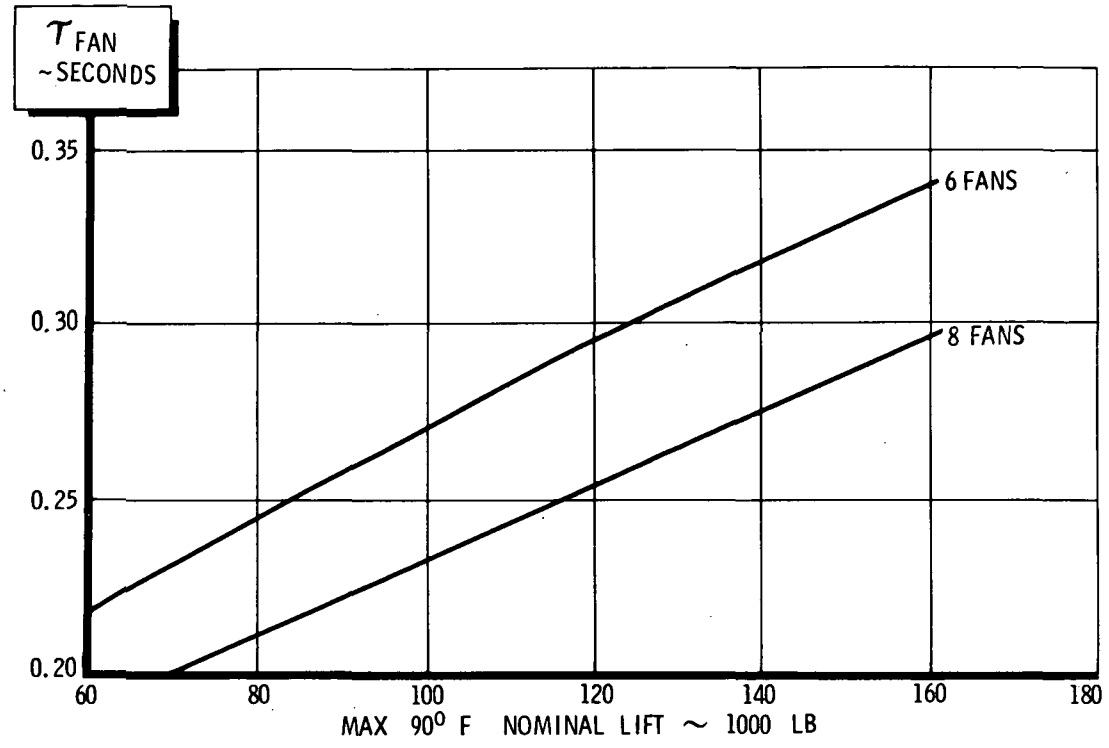

Figure 42. Fan Time Constants for Six and Eight Fan Systems

larger fans required in a 6 fan system required to produce a given total vehicle nominal lift will respond about 0.04 seconds slower than the smaller fans in an 8 fan system. All 8 fan systems indicated would meet the τ_{sys} requirements. Six fan systems will meet the response time requirements with only moderate amounts of spoiling and control quickening for vehicles requiring total installed nominal lifts of 140,000 pounds or less. Above 140,000 pounds the amount of thrust spoiling required would likely begin to severly affect the vehicle propulsion sizing required to maintain the required nominal lift.